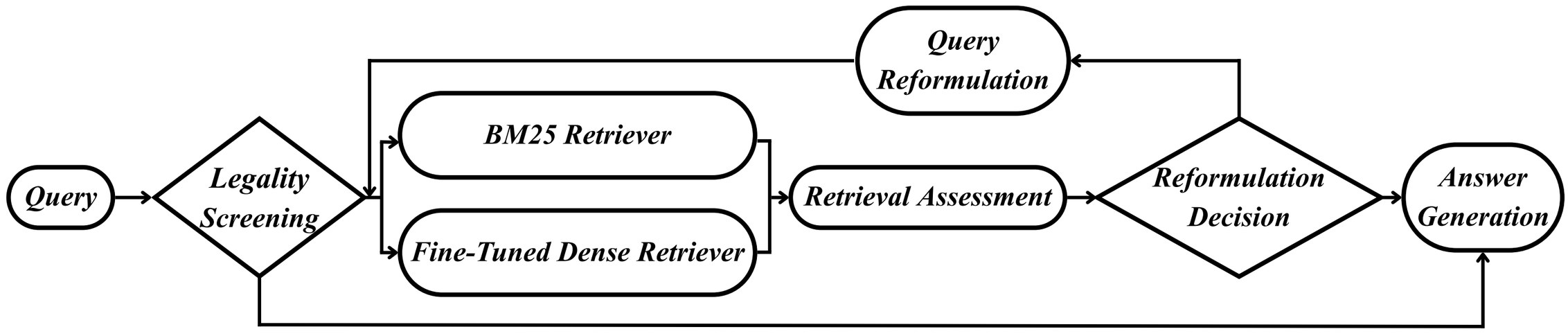
Legal Case Retrieval: An RAG-Based Legal Assistant

Su, Ting-Jui
Department of Computer Science and Engineering

Abstract

Traditional legal case retrieval relies heavily on keyword matching, which often struggles with semantic variation and complex legal queries. This project proposes a ***RAG-based legal assistant*** that combines ***BM25***, a ***fine-tuned dense retriever***, and an ***agentic multi-step workflow*** for query rewriting and evidence assessment. Experimental results show improved ***correctness*** and ***faithfulness*** over both LLM-only and baseline RAG settings. The project demonstrates how hybrid retrieval and evidence-grounded generation can improve reliability in domain-specific question answering.

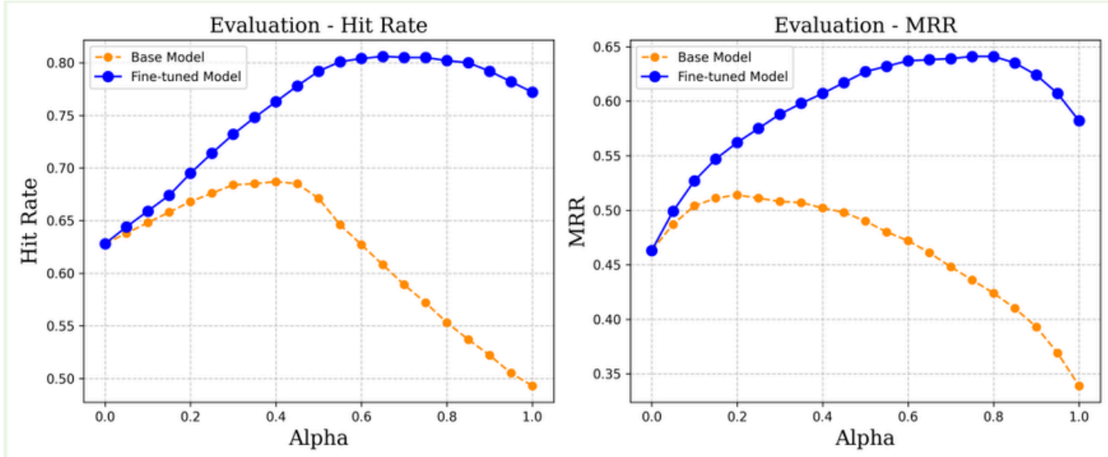
Architecture of the Proposed RAG Framework



Methodology

- **Data Collection and QA Construction:** Judicial Yuan rulings were collected as the legal corpus. An LLM was used to generate a synthetic ***QA dataset*** for retriever fine-tuning and downstream evaluation.
- **Dense Retriever Fine-Tuning:** A multilingual embedding model was fine-tuned using ***contrastive learning*** on the QA pairs. This improved ***Hit Rate@10 by 0.23*** over the base model.
- **Hybrid Retrieval and Agentic RAG:** The system combined ***BM25*** with the fine-tuned dense retriever in ***Weaviate***. Built with ***LangGraph***, the workflow assessed retrieval sufficiency, rewrote queries when needed, and generated answers from accumulated evidence.
- **Evaluation:** System performance was evaluated using ***correctness*** and ***faithfulness***. The proposed agentic RAG framework achieved the strongest overall performance among the compared settings.

Results and Evaluation



System Configuration	Correctness	Faithfulness
LLM-Based (Only)	0.86	
Naïve RAG (Baseline)	0.96	0.89
Agentic RAG (Proposed)	1.00	0.99

Alignment with the SDGs:

SUSTAINABLE DEVELOPMENT GOALS

16 PEACE, JUSTICE AND STRONG INSTITUTIONS

Project Takeaway

This project demonstrates my ability to design and evaluate a domain-specific retrieval pipeline by combining ***hybrid search, embedding fine-tuning, and evidence-grounded generation***. These skills are highly relevant to data science and machine learning, particularly for building reliable retrieval systems, improving semantic representations, and supporting robust real-world decision-making.